PHASE 1 · DATASET VALIDATION · CFM

State Nonfarm Payrolls

The dataset, its clocks, and every validation check

Monthly state-level total nonfarm payrolls (CES, seasonally adjusted) from FRED/ALFRED · source notebook: nfp_validation_new.ipynb

51	**437**	**11,796**	**11,133**	**22**
states	months of history	archived first prints	point-in-time rows	validation checks

Registered results: **11 PASS** **4 WARN** **0 FAIL**

July 2026

One report, two datasets: how state payrolls reach the world

- **What it is** — `{ABBR}NA`: state total nonfarm payrolls. Monthly, **seasonally adjusted**, a level in **thousands of jobs**, from the BLS **Current Employment Statistics (CES)** establishment survey, served by FRED with archival vintages in ALFRED. NSA (`{ABBR}NAN`) rides along for the seasonality contrast; national **PAYEMS** is the benchmark.
- **H1 · Publication schedule** — published in the BLS *State Employment and Unemployment* report, the **same monthly release as state U3**, on a **Friday ~3 weeks after** the national Employment Situation — about **50 days after the reference month**. Verified in V3.2–V3.4.
- **H2 · Reference period** — the figure counts employees on payrolls for the **pay period including the 12th** of the month; FRED stamps the observation on the **1st**. Verified in V2.1.
- **H3 · Revision structure** — CES is revised in the **two months after first release** (sample completion), then **benchmarked annually to the QCEW near-universe count** — the winter benchmark restates roughly the prior 21 months. Current data is a poor proxy for what was known in real time. Quantified in V4.

Note — NFP is the far end of the revision spectrum: claims ~3% revised, U3 86% (but quantized), **payrolls 99% — revised by design**. For this dataset the vintage/current distinction is not a nicety: it is the difference between the number a model could have traded on and a number that did not exist yet.

Three tables, three clocks — same numbers, different questions

table	file	rows	key (grain)	span	question it answers
vintage (df_v)	nfp_vintage_ first_release.csv	11,796	(state, obs_date)	obs 2005-06 → 2026-05 released 2005-07-22 → 2026-06-23	“What was **first published** for month t, and **when**?”
current (sa_c)	nfp_current_ history.csv	22,287 = 51 × 437 months (the 'sa' slice; 'nsa' stored alongside)	(state, obs_date)	obs 1990-01 → 2026-05	“What does FRED say **today** about month t?” (fully revised)
pit	nfp_pit_features.csv	11,133 = ramp 7→51 states × 239 as-of months	(as_of_date, state)	as-of 2006-08 → 2026-06	“What did we actually **know on day d**?” (backtest feed)

Note — The same physical number lives in all three tables — what differs is **when you are allowed to see it**. `df_v` runs on the **information clock**, `sa_c` on the **observation clock**, and `pit` replays both so that **a backtest can only touch what was known at the time**. With 99% of payroll prints later revised — and the QCEW benchmark rewriting 21 months at a stroke — the current table is the single most dangerous input a backtest could take.

Column dictionary — vintage & current tables

column	table	meaning
state / abbr / fips	both	full state name, USPS code, FIPS code (51 incl. DC)
obs_date	both	reference month, stamped on the **1st** — the count covers the pay period including the 12th (V2.1)
adjustment	current	'sa' = seasonally adjusted (validated) · 'nsa' = not adjusted (auxiliary, same file)
value_current	current	today's fully-revised payroll level for that month (thousands)
first_release_date	vintage	day the value **first entered the ALFRED archive** — the information clock
value_first_release	vintage	the first print (thousands) — what a real-time observer saw
is_current	vintage	True if the first print still equals today's value (**~1%** of rows — V4.1)
pub_lag_days	vintage	first_release_date – obs_date: **45–101d, median 50d** (V3.3)

Note — If you remember one thing: **obs_date says what the number measures; first_release_date says when anyone could know it.** For payrolls the gap between the two versions of a month is not just timing — after the QCEW benchmark the **value itself** is different for essentially every month.

Column dictionary — the point-in-time panel (20 columns)

column(s)	meaning
as_of_date	the **vantage point**: a month-start date; every cell in the row reflects only data released **on or before** it (239 consecutive month-starts)
state	one row per state per as-of date (ramps 7 → 51 as states enter the archive — see V3.1)
nfp_t0 … nfp_t12	the **13 most recent monthly payroll levels available** as of that date; t0 = newest (thousands)
nfp_latest_obs	which observation month t0 refers to
nfp_lag_months	as-of month – latest obs month: how **stale** t0 is (median 2, max 4 — V3.7)
nfp_mom1_pct / nfp_mom3_pct / nfp_yoy_pct	1-month, 3-month and 12-month **% changes**, computed only from released readings

Discussion — **“Vantage point” in plain English:** as_of_date is the day you pretend it is *now*. Standing on the 1st of month M, the newest state payroll you legally have is month **M–2** (the ~50-day lag). Crucially, t0…t12 hold the values **as they stood then** — pre-benchmark — not today's rewritten history. Zero look-ahead, verified row-by-row in V3.7.

Why the three tables start on different dates

layer	starts	why it can't start earlier
current history	**1990-01**	FRED serves today's vintage across the whole history — no point-in-time obligation; state CES reaches back to 1990 for every state
vintage (first prints)	**2005-06** (first release 2005-07-22)	ALFRED began archiving state payrolls in Jul-2005 — and only for a **7-state pilot** (MO, MS, KY, IN, TN, IL, AR). The other **44 states join 2007-07-20**. A first print never snapshotted cannot be recovered (V3.1)
pit panel	**2006-08**	a row needs 13 monthly readings (t0…t12) → first valid as-of ≈ 14 months after a state's archive start; the full 51-state cross-section exists from **mid-2008**

- The ordering **current** ≥ **vintage** ≥ **pit** is structural — the same staircase as claims (1986 → 2010-09 → 2010-11) and U3 (1976 → 2005 → 2006). U3 and NFP share the archive boundary exactly, because they ship in the same report.
- Consequence: **1990 → 2005 is NOT point-in-time.** Usable for trend and seasonality estimation; never for backtesting (V3.1).

Note — Reading `describe()` output: the **50% row is the midpoint of the sample, not a boundary** — the pit panel's median as-of lands in 2017 only because the panel spans 2006→2026 and the 7-state early years contribute few rows. The *start* is the min row: 2006-08 (thin), full 51-state panel from 2008–09.

22 checks in 4 modules — how to read the rest of this deck

module	theme	checks	registered results
V1	Structural integrity — states, keys, nulls, positive levels, digit-error guard, truncation	V1.0 – V1.5 (+ missing-value heatmaps)	print-style flags (OK)
V2	Calendar & frequency — the **observation clock** (the month being measured)	V2.1, V2.2a, V2.2b, V2.3	4 PASS
V3	Point-in-time & release calendar — the **information clock** (when it became known)	V3.1 – V3.7	4 PASS · 3 WARN
V4	Revision behavior — first release and after (in % of first print)	V4.1 – V4.4	3 PASS · 1 WARN

- **PASS** = criterion met · **WARN** = usable with a documented caveat · **FAIL** = blocker. This run: **11 PASS, 4 WARN, 0 FAIL.**
- Every check states its **question and criterion up front**; each slide shows the evidence, the verdict, and — where it matters — what it means for modeling.
- Scope: this notebook is the per-question restructure of modules **V1–V4**; deeper modules (outliers, coherence, seasonality, lead-lag, backtest hygiene, governance: V5–V10) live in the full suite `nfp_validation.ipynb`.

MODULE

V1

Is the panel physically sound — right states, unique keys, complete fields, credible payroll levels?

Structural integrity

check	question it answers
V1.0	What do the three tables actually look like? (eyeball before testing)
V1.1	Do we have all 51 states in every table, every period?
V1.2	Is (state, month) a unique key — no double-loaded rows?
V1.3	Are the key fields complete (no nulls)?
V1.4	Is every payroll level finite and strictly positive?
V1.4b	Are the levels a plausible employment magnitude (digit-error guard)? Plus: the missing-value heatmaps
V1.5	Did any state's fetch silently truncate?

V1.0 — The three tables at a glance

INFO

Criterion: look before testing — a validation suite that never prints its data validates blind.

- **vintage**: 11,796 rows · first prints **249k** → **18,181k** — the max is California (~18.2M jobs), the min Wyoming · pub_lag 45–101d, median 50d.
- **current**: 22,287 rows = 51 × 437 months · obs **1990-01** → **2026-05** · revised levels span **196k** → **18,166k**.
- **pit**: 11,133 rows · 239 month-start as-of dates · 20 columns of look-ahead-free features.

	rows	level range (k)	dates
vintage	11,796	249 .. 18,181	2005-06 .. 2026-05
current	22,287	196 .. 18,166	1990-01 .. 2026-05
pit	11,133	(t0..t12)	2006-08 .. 2026-06

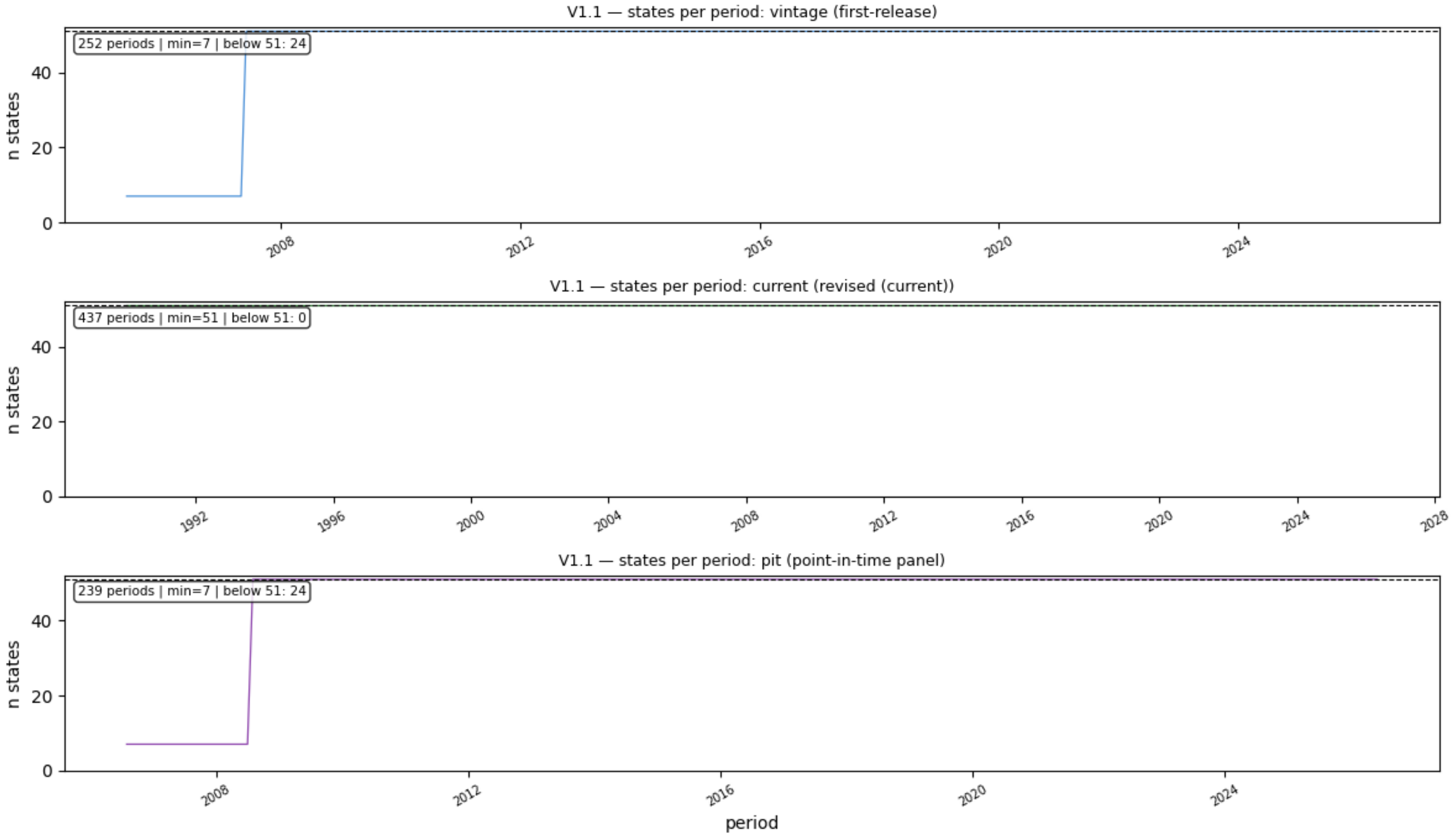
Note — Levels span **two orders of magnitude** across states (WY ~0.3M vs CA ~18M) — cross-state features must be in % changes or shares, never raw thousands. Within a state the level is slow-moving and trending: the natural units are mom/yoy % (exactly what the pit panel precomputes).

V1.1 — Do we have all 51 states in every table, every period?

OK

Criterion: 51 unique states per table; 51 per period outside documented edges.

- Unique states: **vintage 51 · current 51 · pit 51.** Per period: current **51 always** (1990→); vintage and pit dip to **7 states for 24 months (2005-06 → 2007-06)** — the archive pilot era, not missing data.



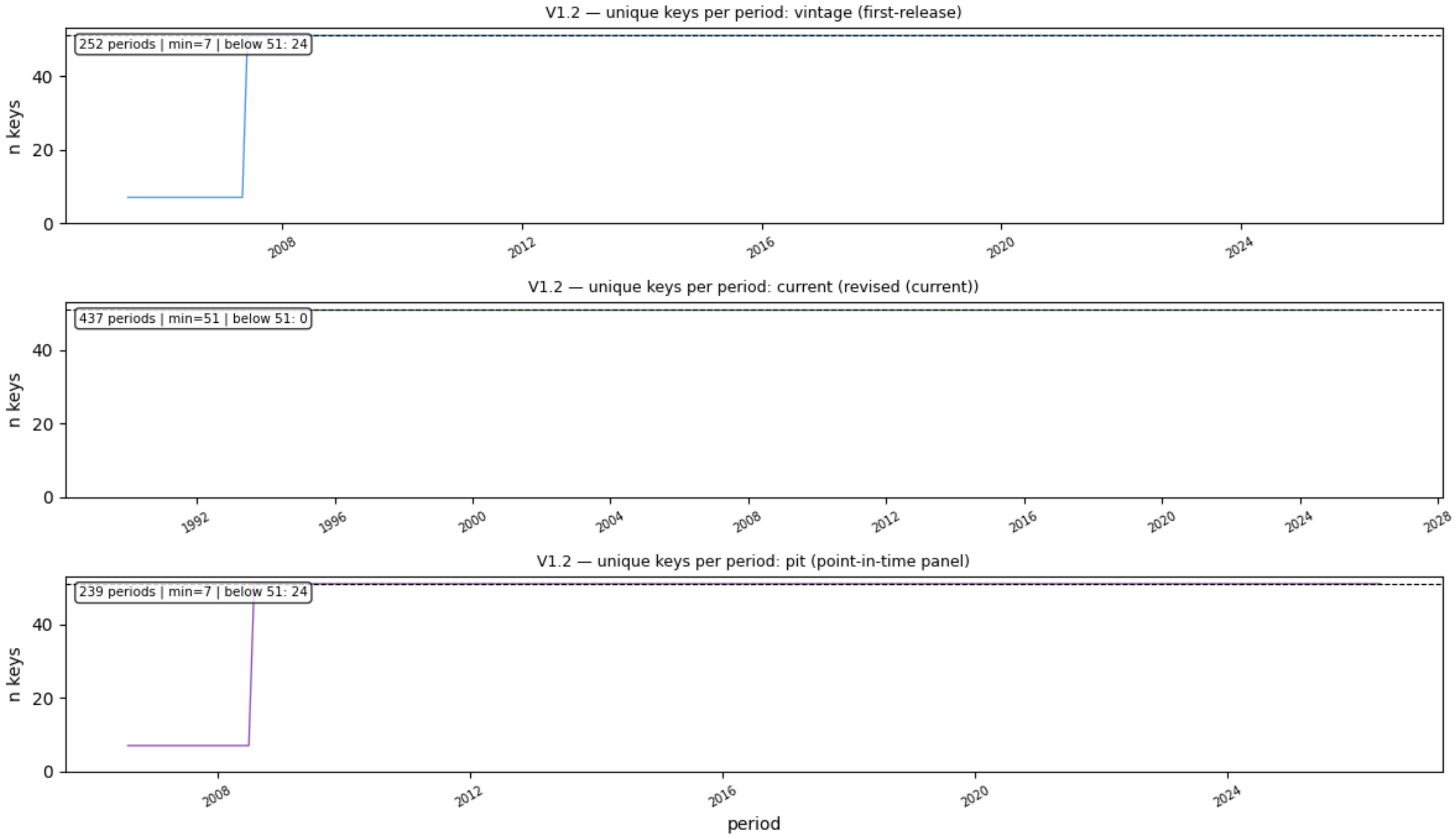
Note — Identical ramp to U3 — the two series ship in the same report and were archived together (same 7-state pilot, same Jul-2007 expansion). The current table is flat at 51 because today's vintage covers all states over the full history.

V1.2 — Is (state, month) a unique key — no double-loaded rows?

OK

Criterion: 0 duplicate rows on the declared key of each table.

- **vintage 11,796 rows = 11,796 unique keys · current 22,287 = 22,287 · pit 11,133 = 11,133** (key: as_of_date × state) — zero duplicates anywhere.



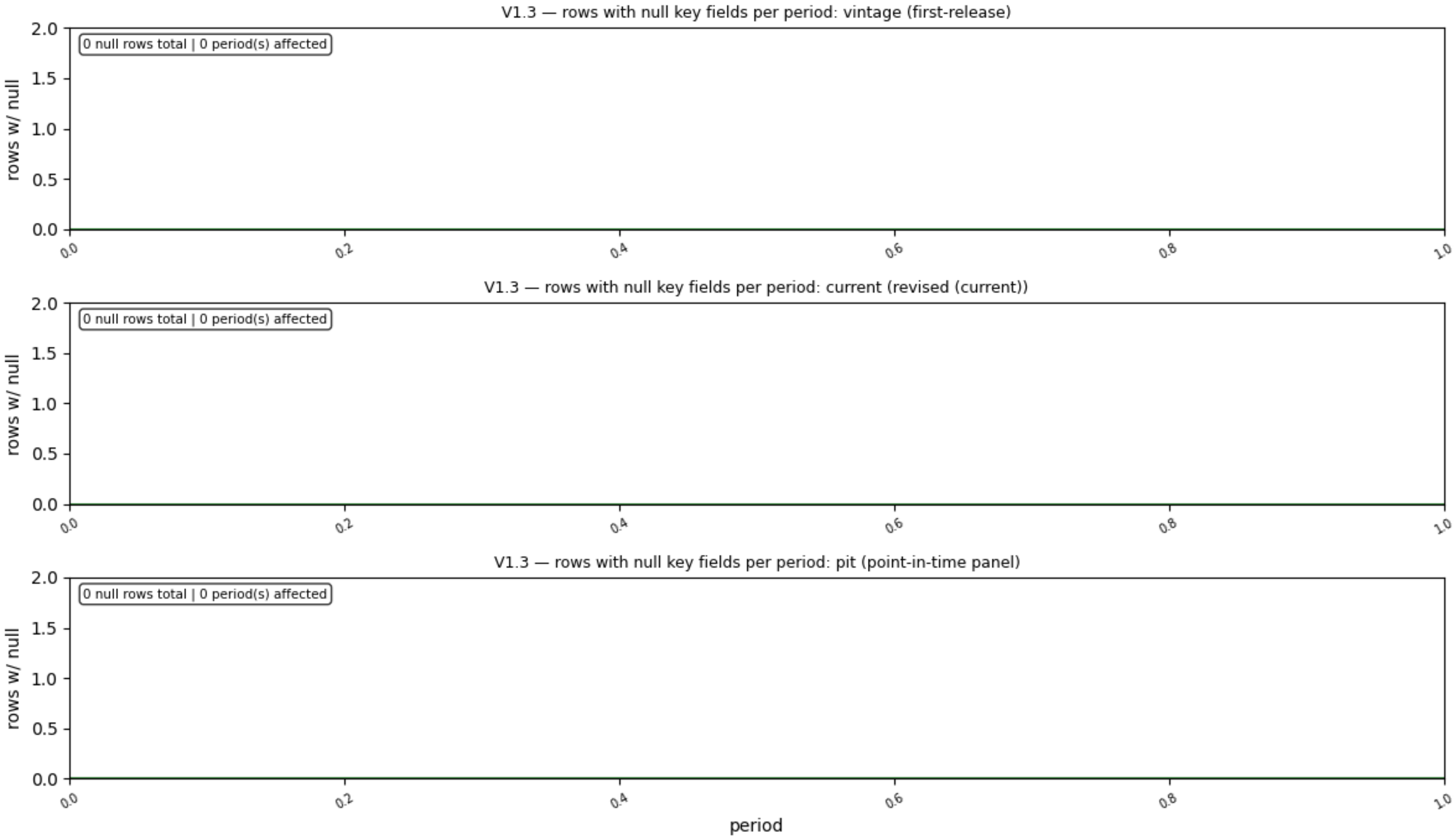
Note — Duplicate keys are the classic symptom of a fetch retried and appended twice. Zero duplicates + V1.5's row-count check together rule out both double-loading and silent truncation.

V1.3 — Are the key fields complete (no nulls)?

OK

Criterion: 0 nulls in every check field of all three tables.

- Field-by-field: **0 nulls** in state, obs_date, first_release_date, value_first_release (vintage); value_current (current); as_of_date, state (pit).



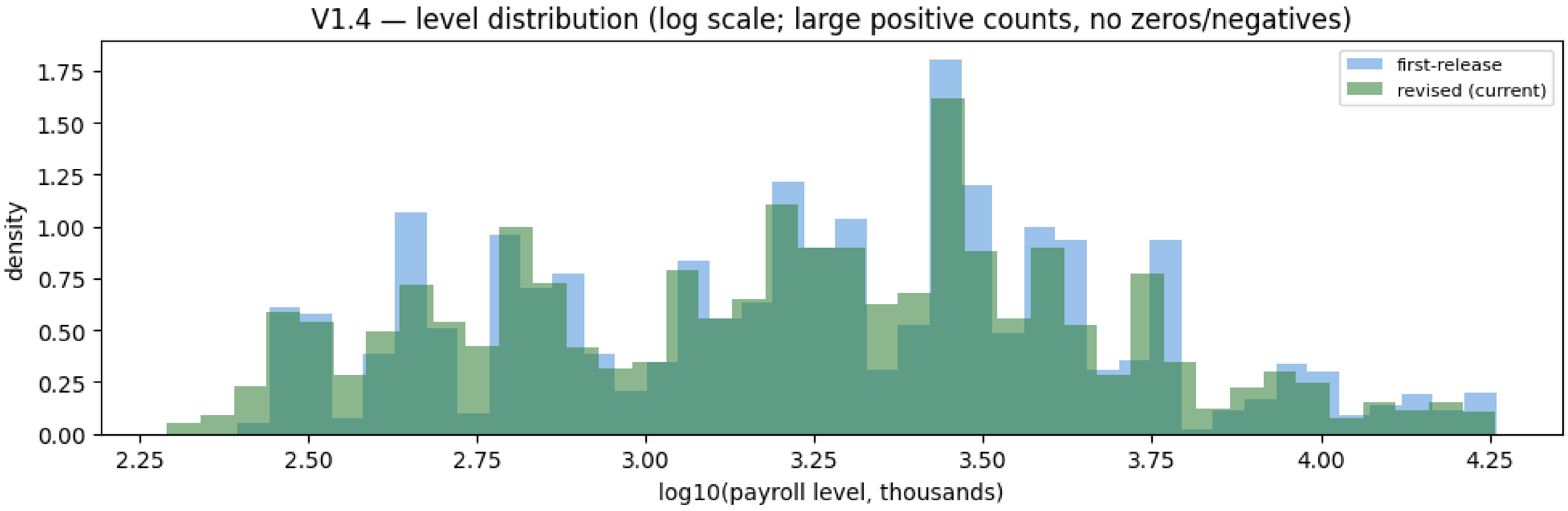
Note — FRED encodes a missing month as the string '.' — the fetch layer converts and drops those, so a null here would mean a parser regression. Note the contrast with U3: payrolls have **no Oct-2025 hole** — the establishment survey kept running through the shutdown; only the household survey (U3's source) was canceled.

V1.4 — Is every payroll level finite and strictly positive?

OK

Criterion: 0 non-finite or non-positive rows.

- **0 bad rows** in both tables. Observed range: vintage **[249, 18,181]k**, current **[196, 18,166]k** — every state, every month, a positive job count.



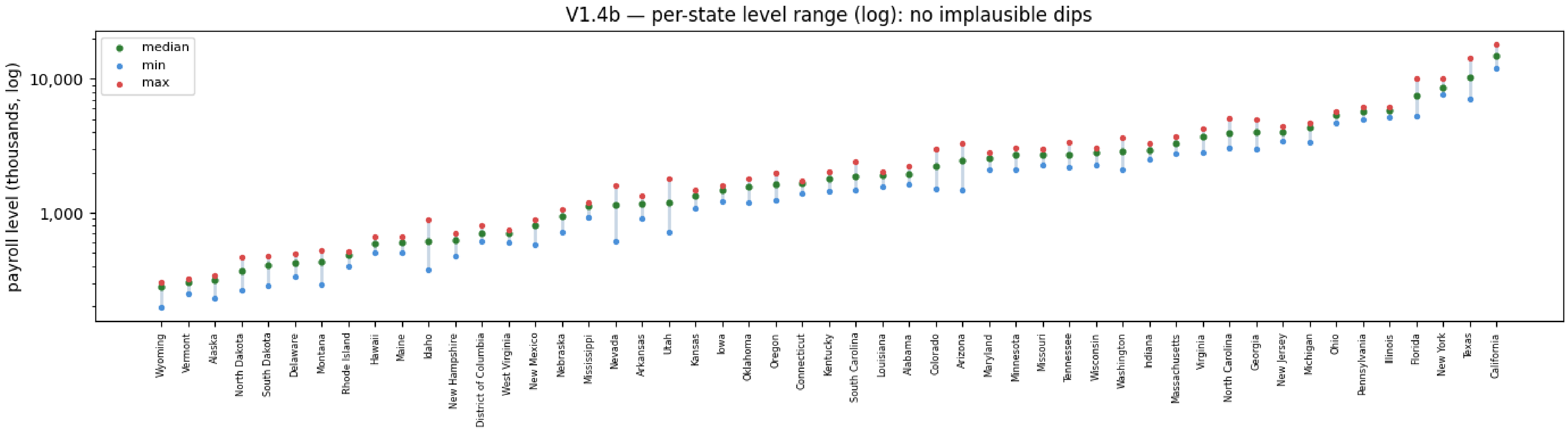
Note — Unlike claims (15 startup-era zeros) there is nothing to flag: CES is a modeled survey estimate, not administrative reporting, so a “zero employment” artifact cannot occur. The interesting magnitude risks are digit errors — V1.4b's subject.

V1.4b — Are the levels a plausible employment magnitude (digit-error / scale guard)?

OK

Criterion: no month below 40% or above 250% of its state median.

- **0 flagged state-months.** The smallest min/median ratio anywhere is **0.53** (a state's early-1990s level vs its median) — comfortably inside the [0.4, 2.5] band.



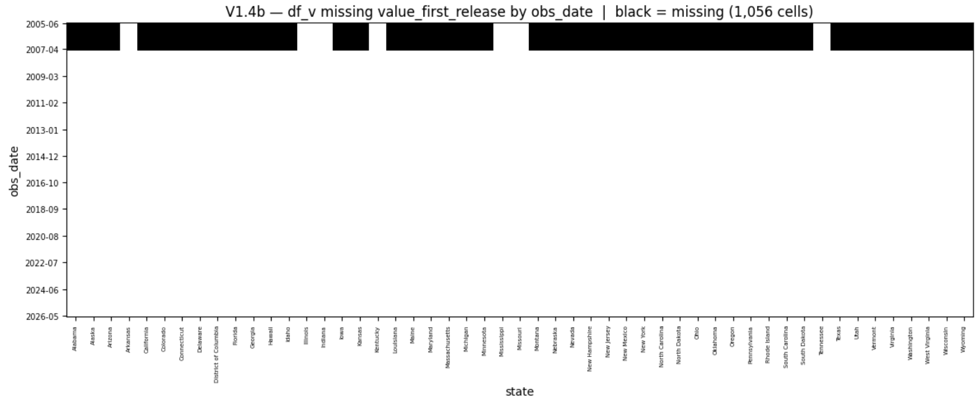
Note — This is the payroll analogue of the claims digit-error hunt (TX +100,000; AZ 132→17,958): a dropped or doubled digit would smash through the band instantly. NFP has **no such error anywhere** — 21 years of vintages, all plausible magnitudes. Even COVID (–15% in Apr-2020) stays far inside a 40% floor.

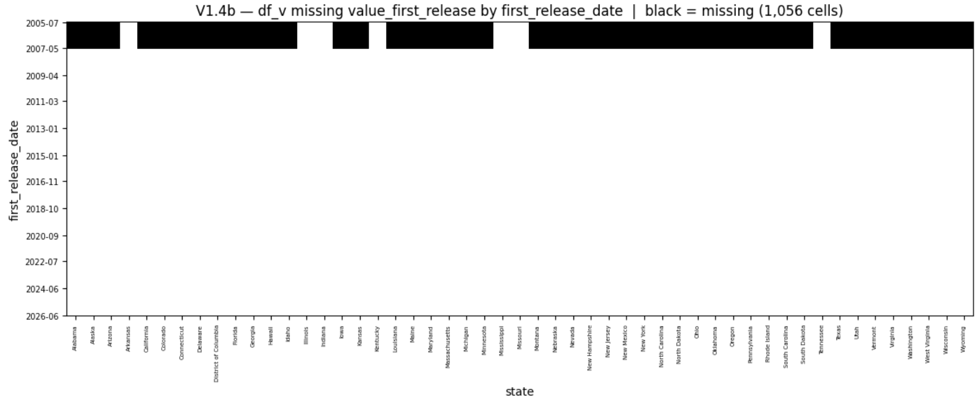
V1.4b (cont.) — Missing-value heatmaps — one block, one explanation

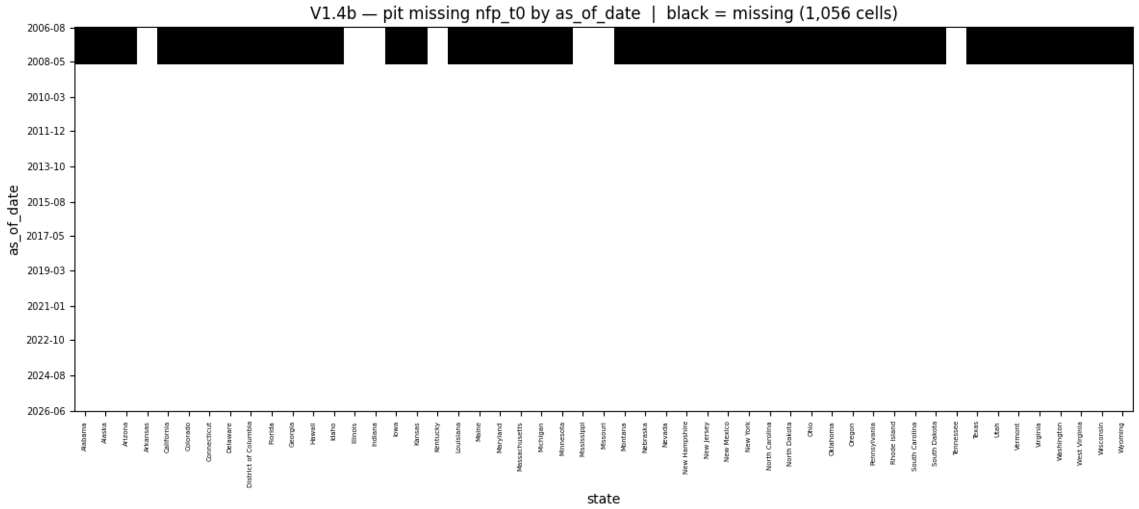
INFO

Criterion: know what a black cell means before calling it missing data.

- vintage by **obs_date** → **1,056 “missing”** of 12,852 (8.22%) · by **first_release_date** → **1,056** (8.28%) · current → **0** · pit nfp_t0 by as_of → **1,056** of 12,189 (8.66%).
- All four numbers are the same object: **44 states × 24 months (2005-06 → 2007-06) = 1,056** — the pre-archive block of the states that joined in Jul-2007.







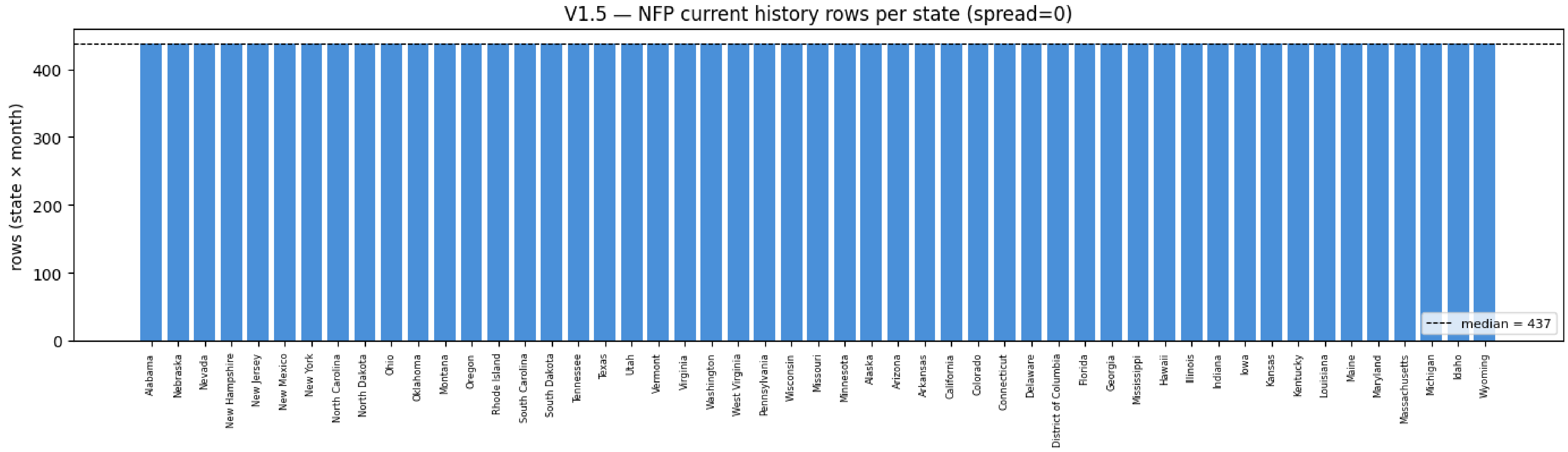
Note — Different story from claims: there, the black cells came from **staggered capture** (post-2020 states archived on different days). Here capture is synchronized — one shared release day per month — so both pivots show the **same ramp-in block**: absence of archive coverage, not data loss inside covered range. The current table shows zero because today's vintage back-fills all history.

V1.5 — Did any state's fetch silently truncate?

OK

Criterion: max – min rows per state ≤ 3 months.

- Rows per state (current history): **437 = 437 = 437** — min, median and max identical, spread **0**. A perfectly rectangular 51 × 437 panel.



Note — State CES publishes every state from 1990-01, so any deviation from 437 would be a fetch failure, not history. (FRED paginates at 100k observations — a truncated series would be conspicuously short.)

MODULE

V2

The observation clock: obs_date stamps the reference month — the count covers the pay period including the 12th.

Calendar & frequency

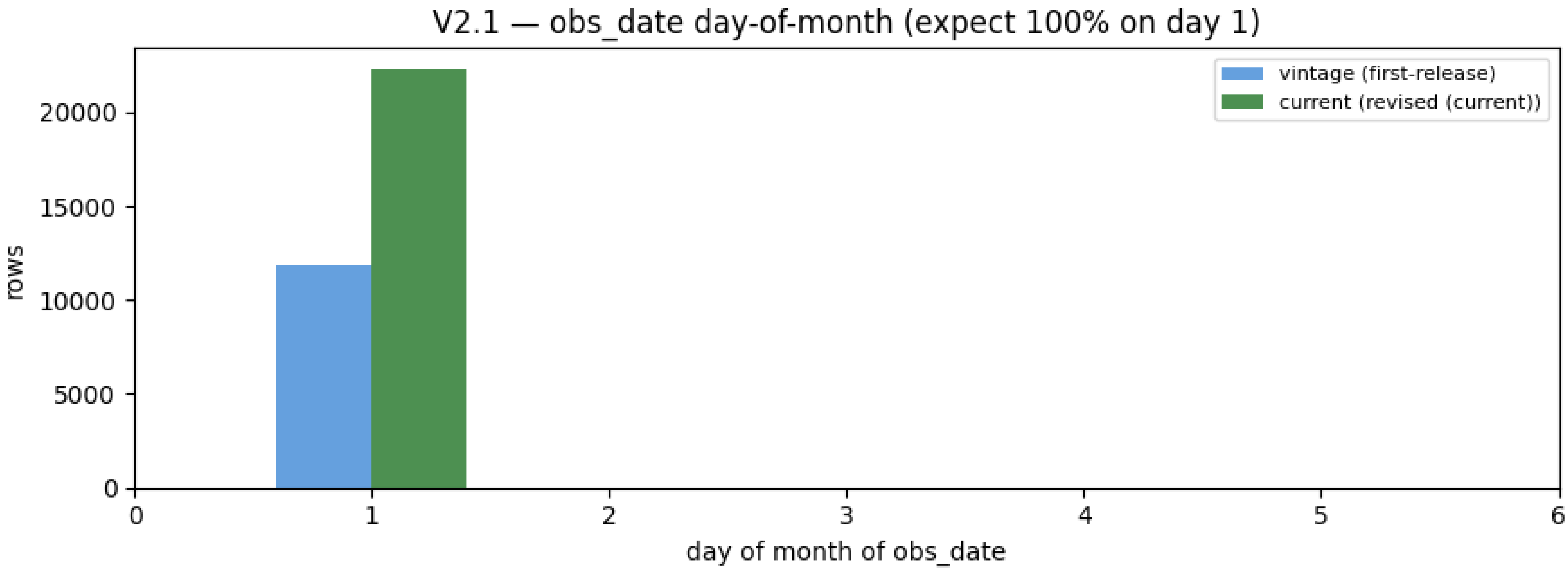
check	question it answers
V2.1	Do all observation dates follow the monthly (month-start) convention?
V2.2a	Are there missing months shared by all states (systematic BLS holes)?
V2.2b	Are there per-state (idiosyncratic) missing months?
V2.3	Do all states cover the same window (balanced panel)?

V2.1 — Do all observation dates follow the monthly month-start convention?

PASS

Criterion: 100% of obs_date have day == 1.

- **0 non-month-start rows** in vintage and current alike.



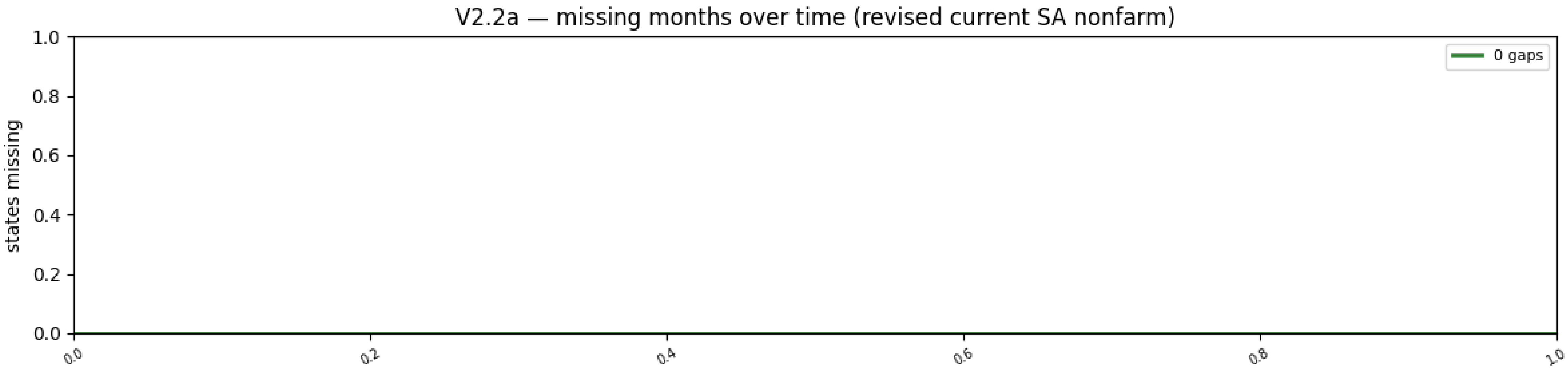
Note — H2 anchors here: FRED stamps the month on the 1st, but the count covers the **pay period including the 12th** — the same reference-week convention as U3 and the national jobs report, which is what makes the three series alignable.

V2.2a — Are there missing months shared by all states (systematic BLS holes)?

PASS

Criterion: documented; shared BLS-wide holes tolerated as WARN.

- **0 systematic missing months** — including **Oct-2025**, which is a hole in state U3.



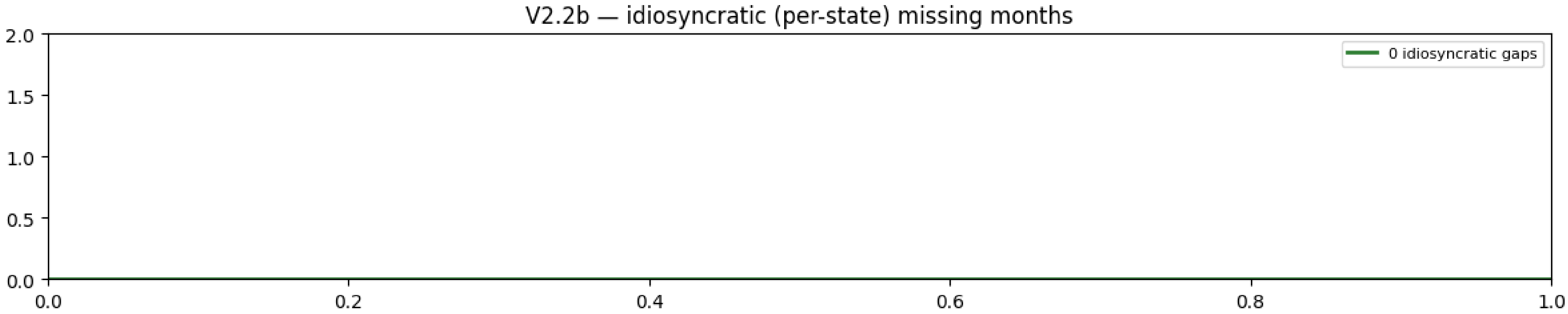
Note — The sharpest cross-dataset contrast in the trio: the late-2025 shutdown canceled the **household survey** (killing Oct-2025 U3 forever) but the **establishment survey** kept collecting — payrolls for Oct-2025 exist. Same report, different survey, different fate.

V2.2b — Are there per-state (idiosyncratic) missing months?

PASS

Criterion: == 0; each occurrence must be investigated.

- **0 idiosyncratic missing months** — no state ever skipped a month on its own in 36 years.



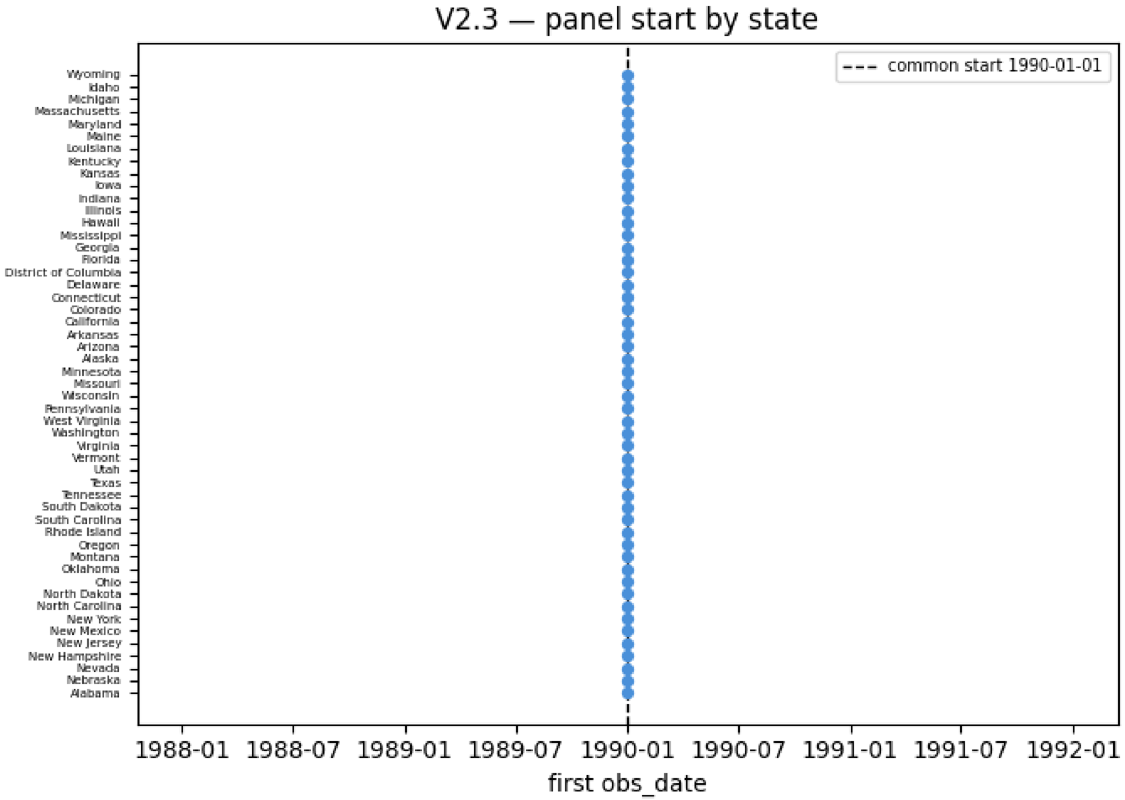
Note — CES is a centrally-produced survey estimate, not state self-reporting — a hole could only be BLS-wide, and there are none. The panel is airtight on the observation clock.

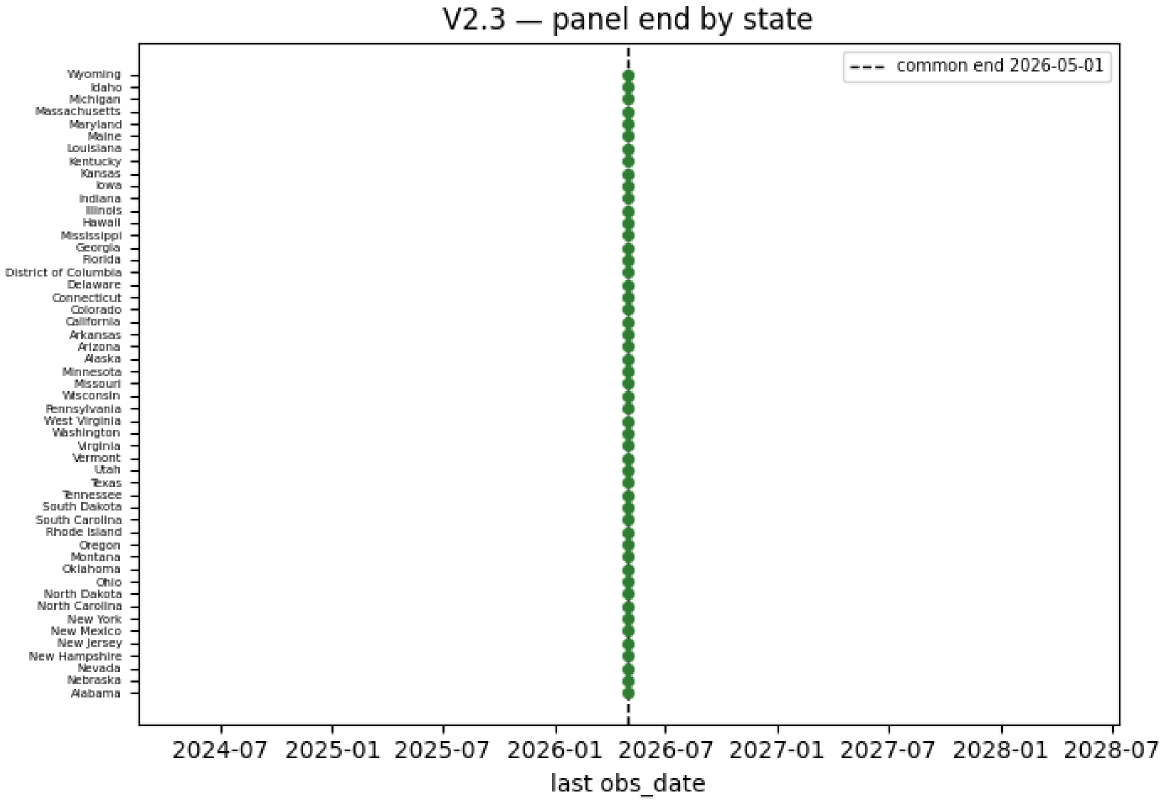
V2.3 — Do all states cover the same window (balanced panel)?

PASS

Criterion: start spread == 0; end spread ≤ 1 month.

- Start spread **0** (every state begins 1990-01) · end spread **0** (every state ends 2026-05). Common balanced window: **1990-01 → 2026-05, 437 months**, no clipping needed.





Note — The joint monthly release keeps all 51 states in lockstep — no live capture edge like the weekly claims panel, because the whole report lands at once.

MODULE

V3

The information clock: first_release_date stamps when the number became known. ~50 days after the month, with state detail 3 weeks behind the national report.

Point-in-time & release calendar

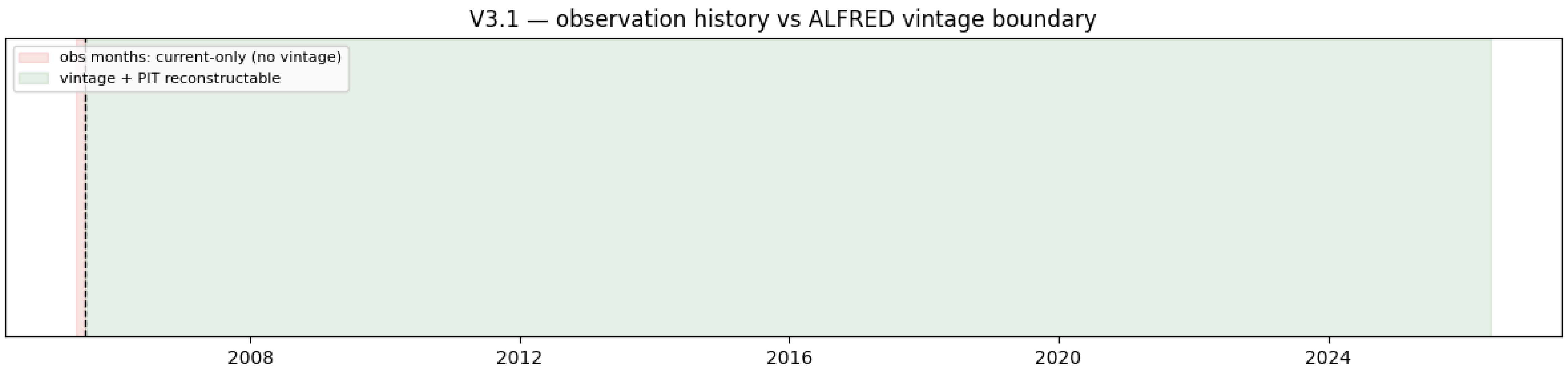
check	question it answers
V3.1	How far back can point-in-time be reconstructed?
V3.2	Does the national vintage calendar match the jobs-report (Friday) schedule?
V3.3	Is the state publication lag ~50 days?
V3.4	Do state releases follow the national jobs report?
V3.5	Is the publication-lag regime stable over time?
V3.6	Do months become available in order (monotone frontier)?
V3.7	Does the PIT panel contain any look-ahead?

V3.1 — How far back can point-in-time be reconstructed?

WARN

Criterion: boundary documented (WARN = limitation, not defect).

- First ALFRED vintage **2005-07-22 — but only for a 7-state pilot** (MO, MS, KY, IN, TN, IL, AR); the remaining **44 states join 2007-07-20**. PIT panel reconstructable from 2006-08 (thin) / **full 51-state cross-section from mid-2008**.



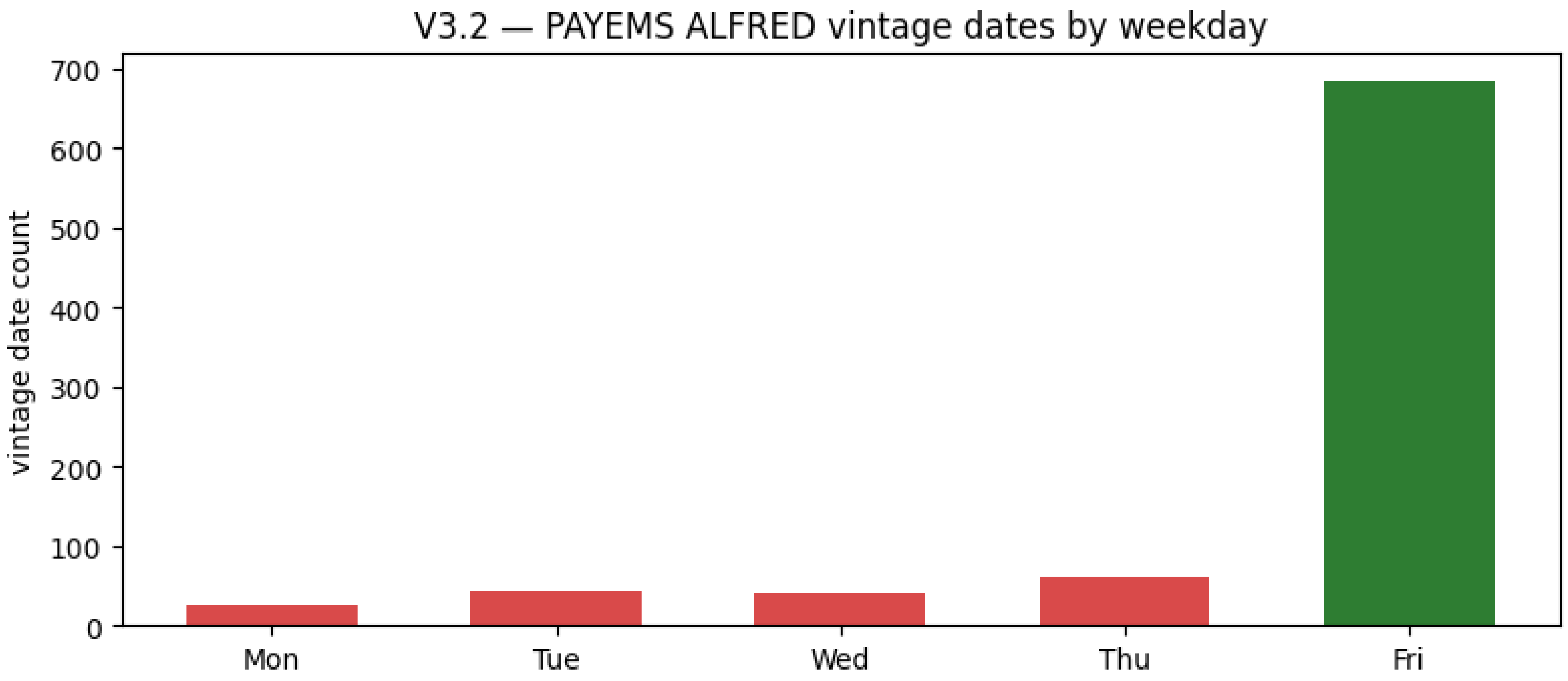
Note — Everything left of the boundary — 1990→2005 — exists only as **today's benchmarked values**: usable for trend and seasonality estimation, never for backtesting. For payrolls this matters doubly: pre-2005 “history” has been rewritten by ~20 annual QCEW benchmarks.

V3.2 — Does the vintage calendar match the jobs-report (Friday) schedule?

WARN

Criterion: Friday ≥ 80% of national PAYEMS vintage dates.

- 857 national PAYEMS vintages: **Friday 79.9%** — a hair under the 80% bar (Mon 3%, Tue 5%, Wed 5%, Thu 7%).



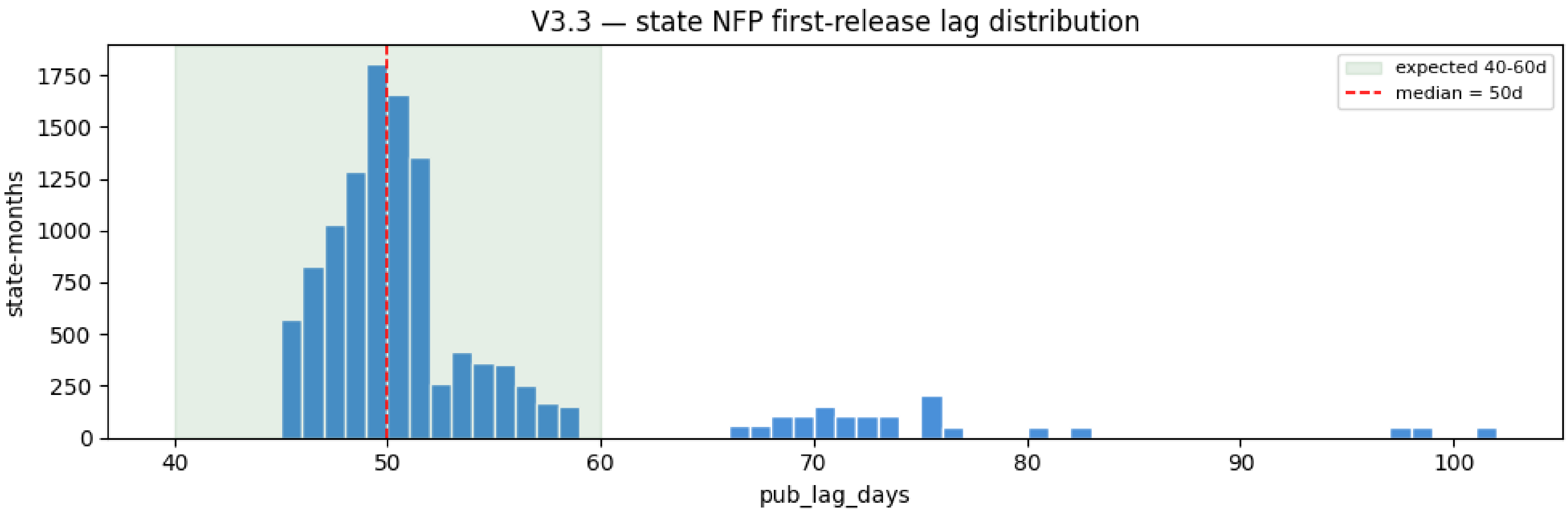
Note — A borderline WARN, not a real anomaly: the Employment Situation itself is a first-Friday release; the non-Friday vintages are ALFRED capture slips and occasionally shifted reports. Treat the release calendar as Friday — but when *timing precision matters*, use the actual archived dates, not the rule.

V3.3 — Is the state publication lag ~50 days?

PASS

Criterion: median publication lag in [40, 60] days.

- 11,796 state first releases: **median 50d, IQR 48–53d**, range 45–101d — about **7 weeks** after the reference month, jointly with state U3.



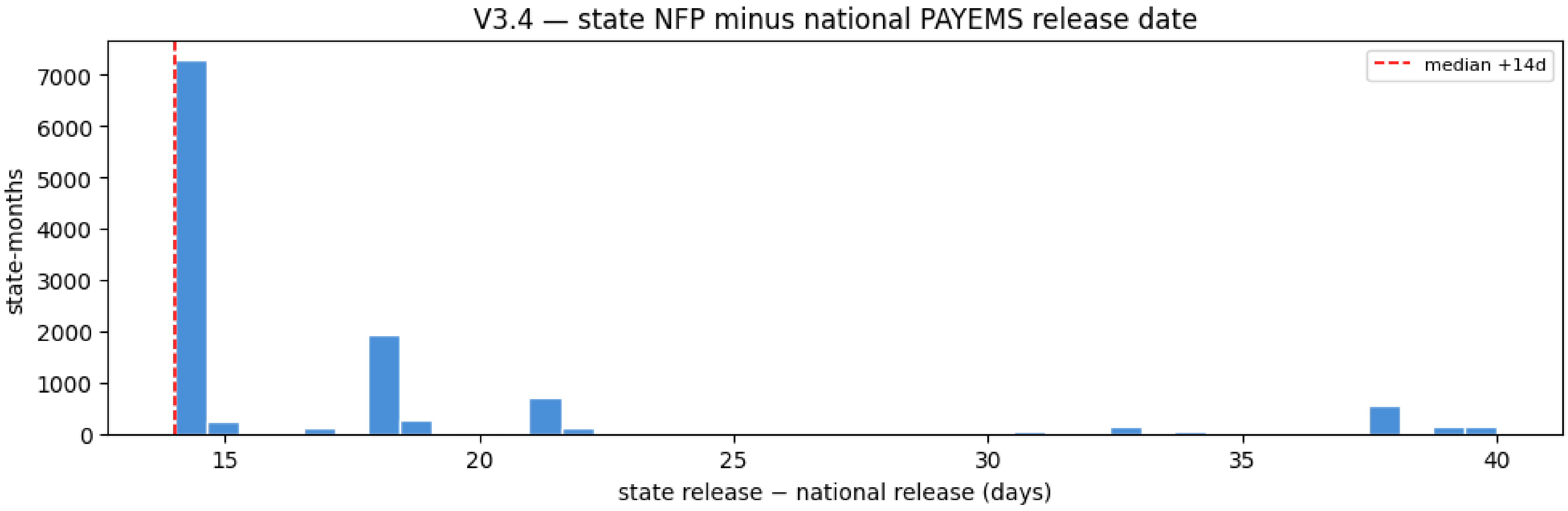
Note — The state jobs number for month M arrives ~6 weeks after the *national* NFP for the same month — and ~7 weekly claims prints covering M are already on the tape. State payrolls are the slowest-moving information in the trio; the nowcasting order is claims → national NFP → state NFP.

V3.4 — Do state releases follow the national jobs report?

PASS

Criterion: ≥ 95% of state releases after the national report for the same month.

- **100.0%** of state first releases land after the national jobs report; **median gap +14d** (IQR 14–18, max 56).



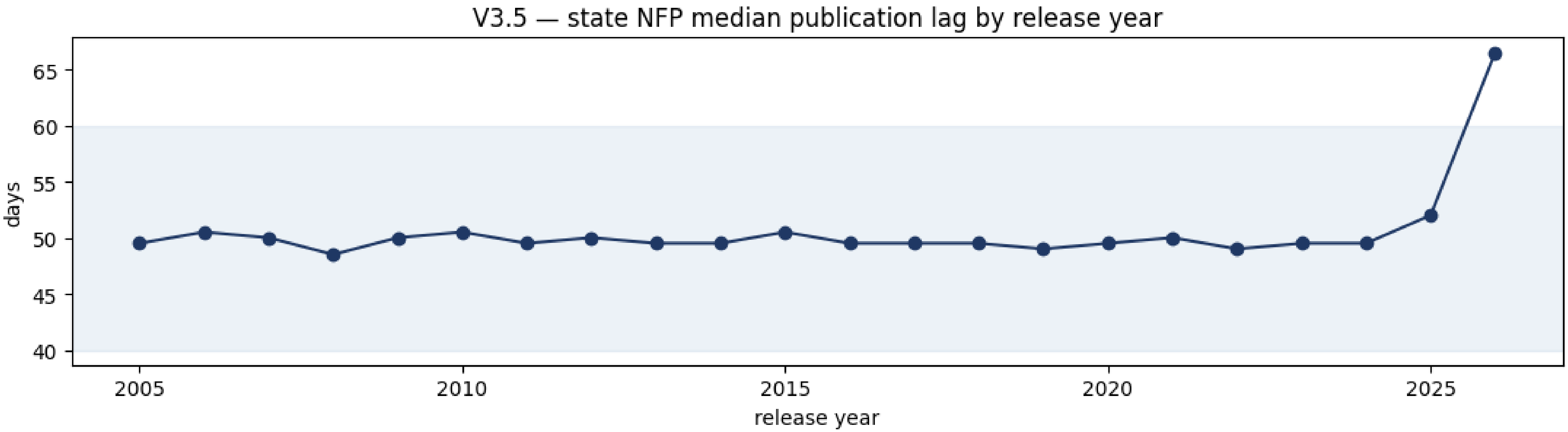
Note — State detail never front-runs the national aggregate — the national print is legitimately in your information set when forecasting the state cross-section two weeks later. For NFP this is the classic setup: national surprise first, state decomposition second.

V3.5 — Is the publication-lag regime stable over time?

WARN

Criterion: yearly median lag range ≤ 7 days.

- 2005–2024: yearly medians pinned in **48.5–50.5d** — remarkably stable. Then **2025: 52d and 2026: 66.5d** blow the range out to 18d — the shutdown-disrupted release calendar catching up.



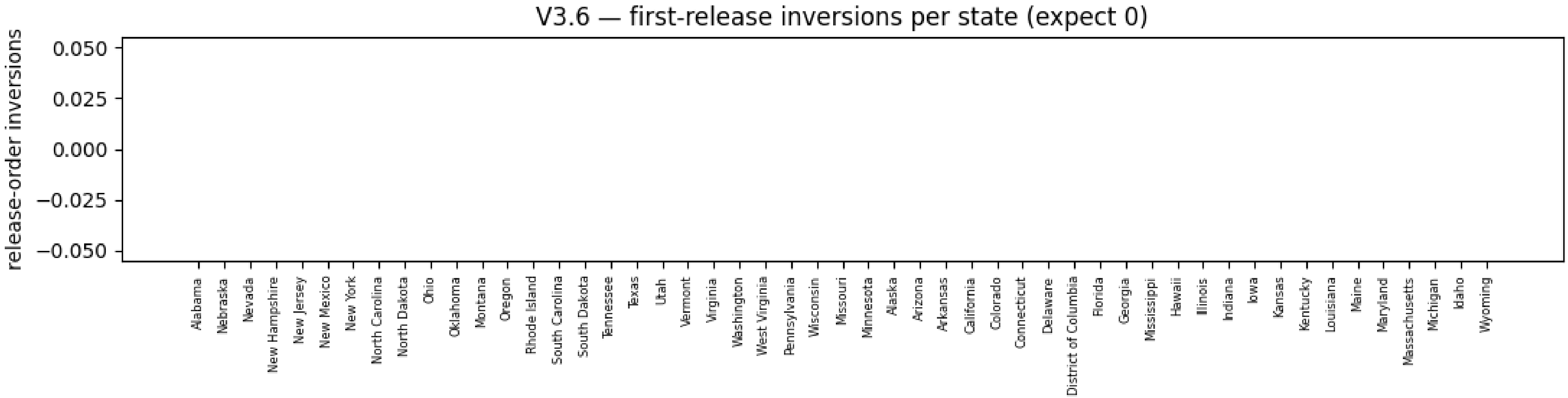
Note — One rule for 19 years, then a live disruption: backtests through 2024 may assume a constant ~50d lag; **2025–26 need actual release dates**, not the rule. Expect renormalization once the delayed reports catch up — identical picture in U3, as the two ship together.

V3.6 — Do months become available in order (monotone frontier)?

PASS

Criterion: 0 release-order inversions per state.

- **0 inversions** — every state's release dates are weakly increasing in observation order.



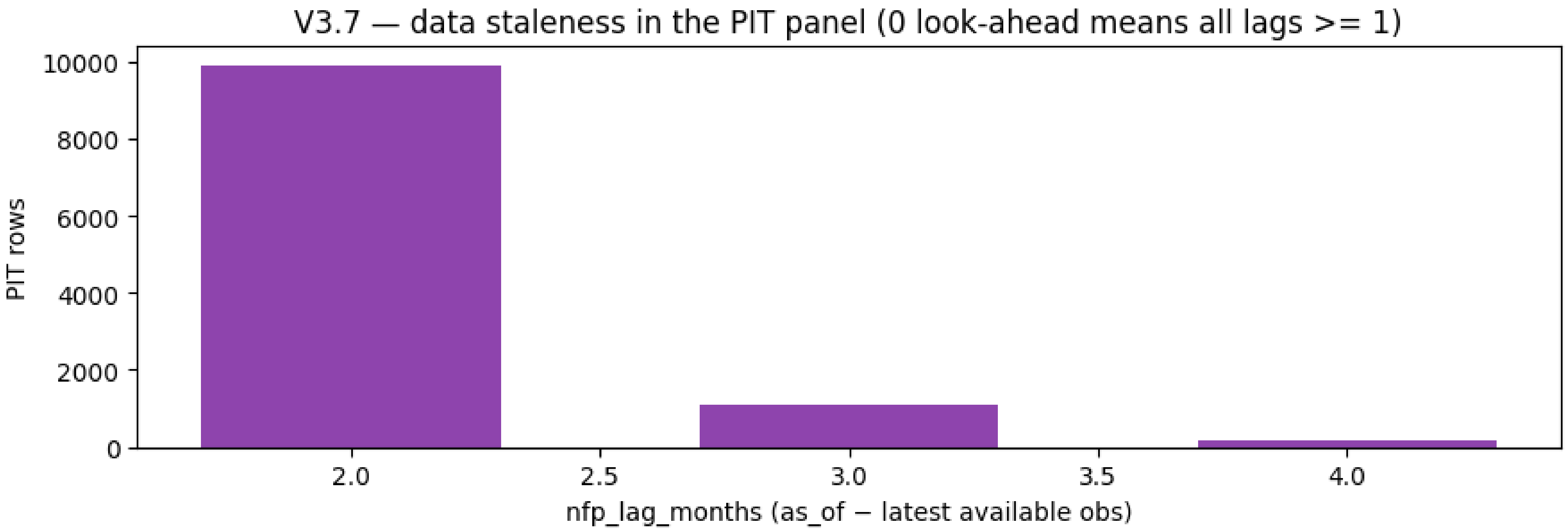
Note — An inversion would mean a later month became visible before an earlier one — poison for “latest available” logic. Monotonicity is what lets the pit builder use a simple cummax **availability frontier** with no correction passes.

V3.7 — Does the PIT panel contain any look-ahead?

PASS

Criterion: 0 violations, 0 unmatched keys.

- Re-verified all **11,133 rows**: release date of the underlying value ≤ as_of_date — **0 violations, 0 unmatched keys**.
- Staleness (nfp_lag_months): **median 2, min 2, max 4** — at any month-start you know state payrolls through roughly two months back.



Note — The hard gate for backtesting, recomputed **against the vintage table** rather than the builder's own bookkeeping — a builder bug could not certify itself. Remember what the cells hold: the values **as first published**, i.e. pre-benchmark — the panel is honest about levels a backtest could have known.

MODULE

V4

The central module for payrolls: revised almost every month, then benchmarked annually to QCEW — measured in % of the first print.

Revision behavior

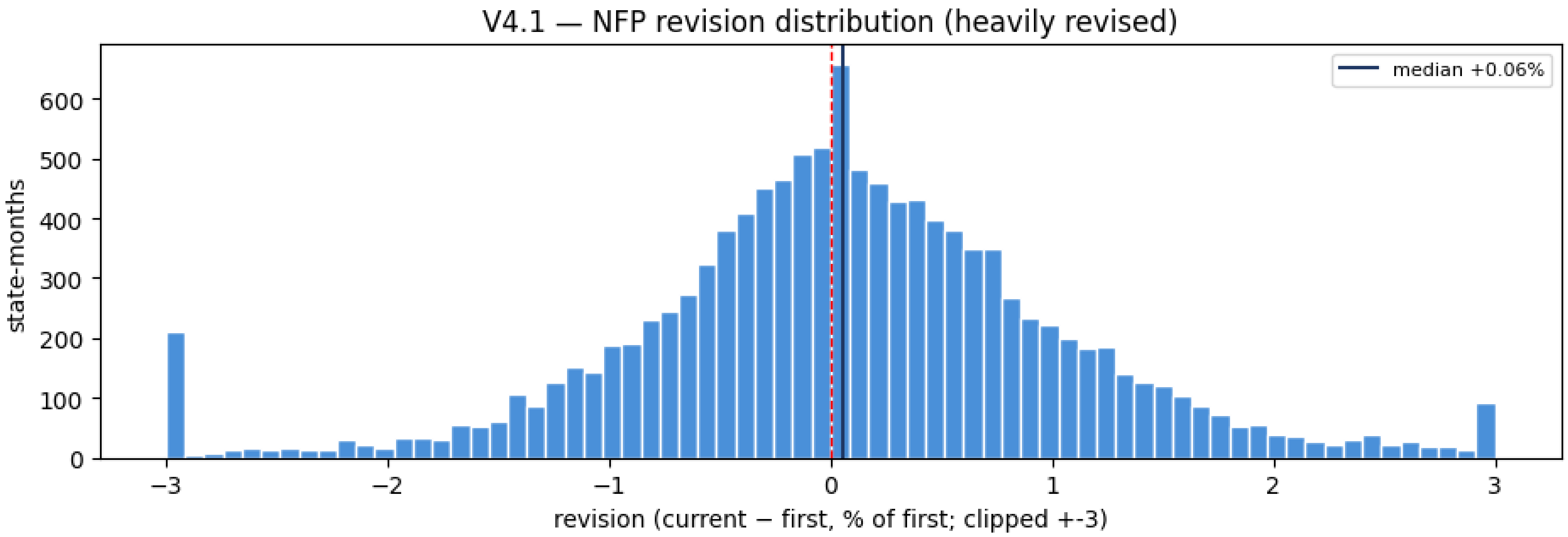
check	question it answers
V4.1	How often, and by how much, does a first print get revised (endpoint: first vs current)?
V4.2	Do revisions cluster at the annual (winter) QCEW benchmark?
V4.3	Is there a systematic bias in first prints?
V4.4	How large is the national PAYEMS revision (calibration)?

V4.1 — How often, and by how much, does a first print get revised?

WARN

Criterion: informational — NFP is heavily revised by design.

- 11,796 obs compared: **99% revised** · median |revision| **0.55%** · p90 **1.63%** · max **7.5%**.
- Largest upward corrections cluster in the **2020–21 recovery** — Nevada Dec-2021 +5.1% (+70.6k jobs), Hawaii and Washington close behind: real-time prints materially understated the rebound.



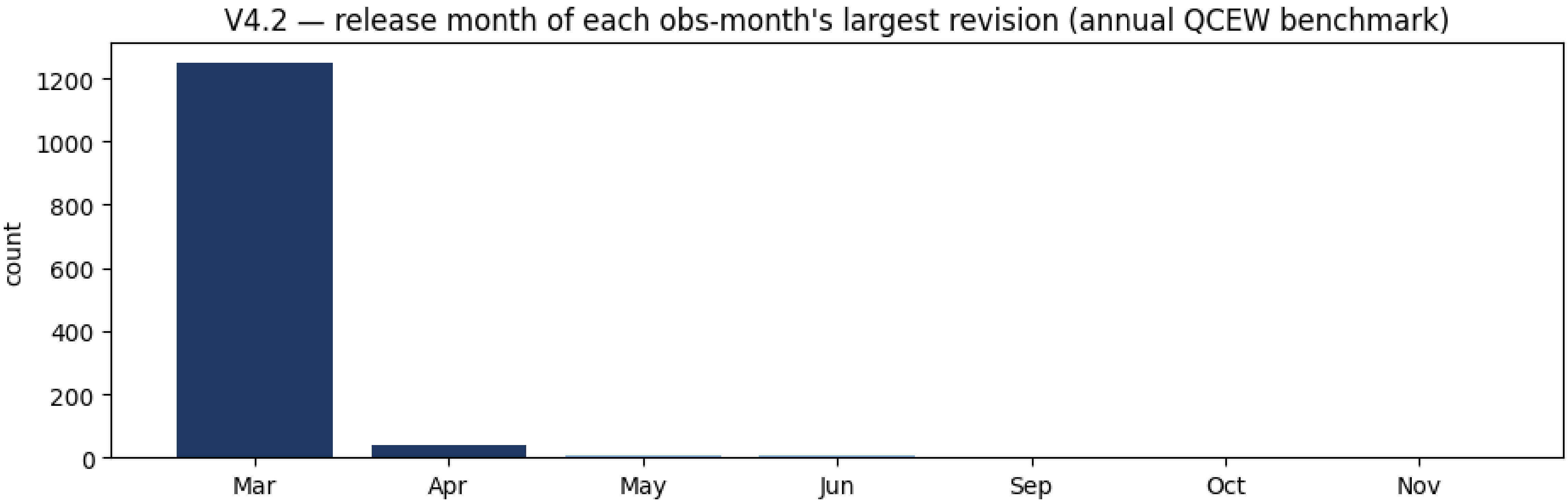
Note — The exact opposite of claims: “what does FRED say today” differs from “what was known then” for **essentially every month**. Any strategy or study fed current payroll history is trading on QCEW benchmarks published up to two years later — vintage data is not optional here, it is the dataset.

V4.2 — Do revisions cluster at the annual (winter) QCEW benchmark?

PASS

Criterion: ≥ 60% of largest revisions released Jan–Apr.

- **99%** of each obs-month's largest revision lands in a **Jan–Apr** release; modal month **March** (1,249 of 1,308 obs; CA/TX/NY full vintage trajectories).



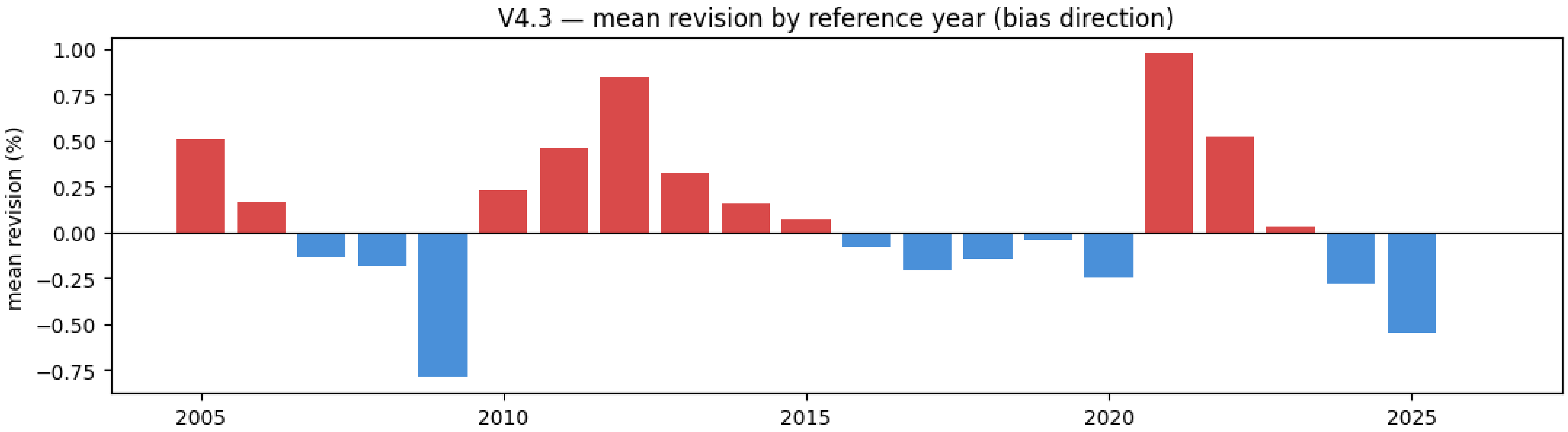
Note — The annual QCEW benchmark re-anchors the survey to the near-universe employment count and restates ~21 months in one pass. Apply revisions **on their release date, never backward** — using post-benchmark values before the following March is look-ahead by two years in the worst case.

V4.3 — Is there a systematic bias in first prints?

PASS

Criterion: |mean revision| < 0.10%.

- Mean revision **+0.052%** (median +0.057%, std 1.11%): first prints **understate** employment slightly on average — the survey's late respondents and new-business births get added later.



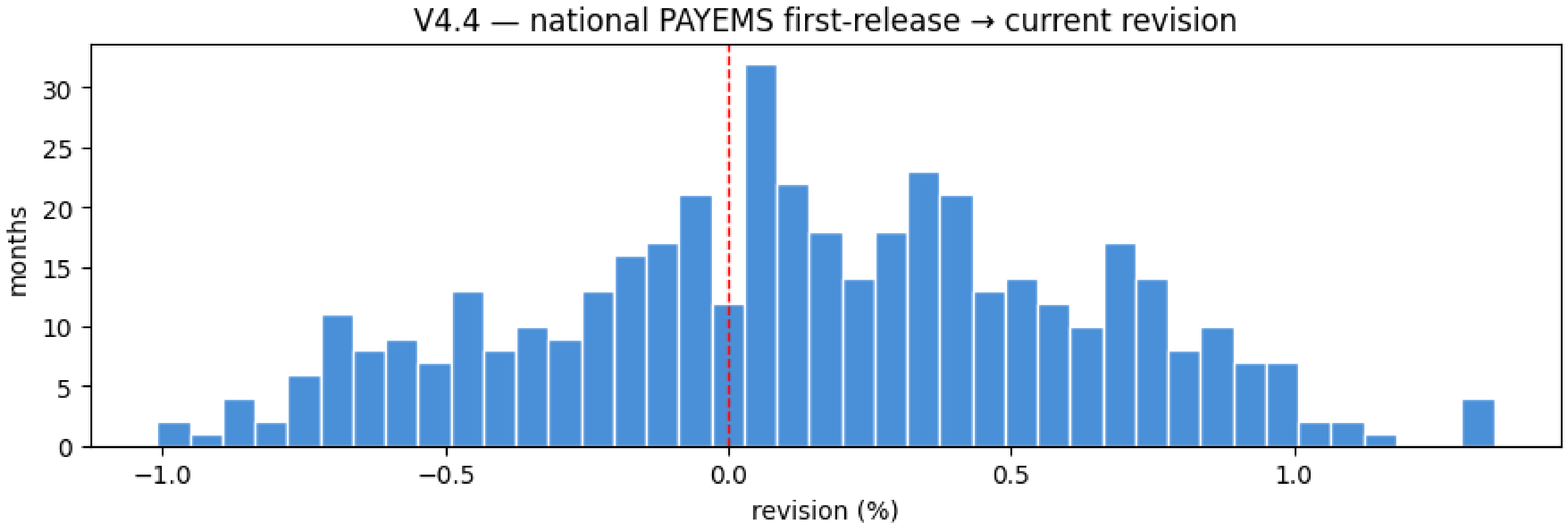
Note — A persistent sign means first prints are **predictably** off — a modellable nowcast signal: expected true level ≈ first print × (1 + 0.0005), with the by-year chart showing when the sign flips (recessions revise down, recoveries revise up).

V4.4 — How large is the national PAYEMS revision (calibration)?

PASS

Criterion: informational calibration on the national series.

- National PAYEMS first-release → current: **median +0.13%, IQR [−0.19, +0.46], max |1.4%|** — the state median (0.55%) is roughly **4× the national**.



Note — Same benchmark machinery, larger sample: the national series is the tamer sibling. If your model consumes both, weight the national print's real-time reliability above any single state's — and expect state-level surprises to partially wash out in aggregation.

Scorecard — every registered check

ID	check	result	headline evidence
V2.1	All obs_date are month-starts	**PASS**	0 non-month-start rows
V2.2a	Systematic (shared) missing months	**PASS**	none — even Oct-2025 present (establishment survey ran)
V2.2b	Idiosyncratic (per-state) missing months	**PASS**	none in 36 years
V2.3	Panel edges aligned across states	**PASS**	start spread 0 · end spread 0 (1990-01 → 2026-05)
V3.1	ALFRED vintage boundary identified	**WARN**	2005-07 (7-state pilot) / 2007-07 (all 51); pre-2005 trend-only
V3.2	National vintages land on jobs-report Friday	**WARN**	Friday 79.9% — 0.1pp under the bar; capture slips, not schedule
V3.3	State publication lag ~50 days	**PASS**	median 50d (IQR 48–53)
V3.4	State releases follow the national report	**PASS**	100% after national; median gap +14d
V3.5	Publication-lag regime stable	**WARN**	48–50d for 19 years; 2025–26 disrupted (52d, 66.5d)
V3.6	Release dates weakly increasing per state	**PASS**	0 inversions
V3.7	PIT panel look-ahead guard	**PASS**	0 violations / 0 unmatched of 11,133 rows; staleness 2–4m
V4.1	Endpoint revision rate & magnitude	**WARN**	99% revised; median |0.55%|, p90 1.63%, max 7.5%
V4.2	Revisions cluster at the QCEW benchmark	**PASS**	99% of largest revisions in Jan–Apr; modal March
V4.3	Systematic bias in first prints	**PASS**	mean +0.052% — first prints slightly understate
V4.4	National PAYEMS revision calibration	**PASS**	median +0.13%, max |1.4%| — states ~4× the national

Module V1 uses print-style flags rather than registered rows: V1.1 OK · V1.2 OK · V1.3 OK · V1.4 OK (all positive) · V1.4b OK (0 magnitude flags — no digit errors anywhere) · V1.5 OK (perfect 51×437 rectangle).

What a model built on this data must respect

- **Vintage data is the dataset.** 99% of first prints change; current history embeds QCEW benchmarks published up to two years later. The pit panel (0 look-ahead, pre-benchmark values, staleness 2–4 months) is the only table a backtest may touch.
- **The information clock is the slowest in the trio.** Median 50-day lag, after the national report (+14d) and after ~7 weekly claims prints — the nowcasting order is claims → national NFP → state NFP.
- **The point-in-time world starts 2005-07 for 7 pilot states, 2007-07 for all 51** — treat mid-2008 as the first clean cross-section. 1990→2005 is trend/seasonality estimation only.
- **Benchmark season is Jan–Apr, modal March.** 99% of the biggest revisions land there; apply them on release date, never backward. First prints carry a small persistent understatement (+0.052%) — a modellable nowcast signal.
- **Scale across states, not within.** WY ~0.3M vs CA ~18M jobs: cross-state features must be % changes or shares; the panel precomputes mom1/mom3/yoy.
- **Structurally the cleanest of the three.** No holes (even Oct-2025 present), no digit errors, perfect rectangle — NFP's entire risk budget is concentrated in one place: **revisions.**

Note — Verdict: **11 PASS · 4 WARN · 0 FAIL** — cleared for forecasting, lead-lag and backtesting under the rules above. The WARNs document the data's plumbing (archive ramp-in, borderline Friday share, shutdown-era lag, by-design revisions), not defects; deeper modules run in the full suite nfp_validation.ipynb as V5–V10.